Week 4 – Decisions Made

1. **TCP sockets were selected as the IPC mechanism** after considering the need for communication between independent Java processes.
2. **Port 5001 was assigned for UI–Core communication and Port 5002 for logging communication** to keep command execution and logging traffic separate.
3. **LinkedBlockingQueue was retained only for internal thread coordination**, while socket communication was used between independent processes.
4. The team decided to **retain the existing HTML-based UI** and adapt its communication layer instead of redesigning the complete interface.
5. The existing CPU, Memory, Stack, and FIFO Queue implementations were **reused inside the Core Process** rather than rewriting the simulator logic for Week 4.
6. **RUN was implemented through repeated STEP requests**, allowing the UI to control continuous execution and stop it without terminating the Core Process.
7. **STATE polling messages were excluded from execution logs** to prevent unnecessary repetitive entries in `simulator.log`.
8. Logging was kept as a **separate process** so that execution records could be maintained independently from CPU execution.
9. The same test program was used for the standalone and multi-process versions to make the **performance comparison more consistent**.
10. **Execution time, memory usage, and CPU usage** were selected as the main metrics for comparing the standalone and multi-process simulator.
11. Development remained divided according to the assigned components, while **final IPC integration, interface resolution, system testing, and benchmarking were handled during integration**.
12. Final acceptance of the Week 4 implementation would be based on successful **UI → Core execution, Core/Logging communication, simulator state updates, log generation, and benchmark testing**.